PIPELINE AUDIT AND HEALTH REPORT

Food and drink: specialty coffee roastery

Three kinds of green coffee share two roasters; switching origin costs a 4-minute cleaning changeover. Roasted lots cool, rest in a degassing silo for 30 minutes, pass a cupping panel and are bagged in sixes. A good example of changeovers: try lowering the setup time or splitting the origins across roasters.

Generated 2026-10-13 19:48 | simulated 8.0 h (warm-up 60 min) | seed 71 | 10 replications

HEALTH SCORE

86 /100

Grade B

Good: sound design with minor weaknesses worth addressing.

Critical 0 | High 1 | Medium 1 | Low 2 | Info 2

The score starts at 100 and loses points for each finding. The deductions are listed in Appendix B.

THROUGHPUT	COMPLETED	ARRIVALS LOST
6.3 units/h	44 units	0.0%

AVERAGE WIP	CYCLE TIME	BOTTLENECK
11.9 units	5863 s	roaster_1

Executive summary

Primary bottleneck: 'roaster_1' (58% busy, 41% down, index 100). Sources are held back up to 13% of the time. Run what-if to measure which lever recovers the most throughput.

What to fix first

- High (roaster_1): Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'roaster_1' (+4.2% throughput).
- Medium (roaster_1): Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.
- Low (throughput): Use a longer run if you need tighter estimates.
- Low (run length): Aim for at least 30x the cycle time so averages settle.

Best measured improvement

Best lever: 10% faster processing at 'roaster_1' -> +0.3 units/hour (4.2%), measured over 3 seeds with common random numbers.

Findings

Each finding names where the problem is, why it matters, and what to do. Severity reflects the impact on throughput, correctness or confidence in the results.

SEVERITY	AREA	WHERE	ISSUE AND RECOMMENDED ACTION	POINTS
High	Capacity	**roaster_1**	The bottleneck is 100% utilised (busy or down). There is no slack: any disturbance directly cuts output. Action: Add parallel capacity, speed the station up, or reduce downtime. Measured best lever: 10% faster processing at 'roaster_1' (+4.2% throughput).	-8
Medium	Reliability	**roaster_1**	Down 41% of the time (3 failures). Action: Shorten repairs (spares, on-call technicians) or introduce preventive maintenance. A 20% shorter repair time is a what-if lever.	-3
Low	Confidence	**throughput**	Throughput 95% CI is +-5.7% of the mean. Action: Use a longer run if you need tighter estimates.	-1
Low	Model	**run length**	The measured window (7.0 h) is only 4x the cycle time. Action: Aim for at least 30x the cycle time so averages settle.	-2
Info	Capacity	**roaster_1**	Constraint of the line: 100% utilised. Action: Improvements elsewhere will not raise throughput until this station is relieved. Measured best lever: 10% faster processing at 'roaster_1' (+4.2% throughput).	
Info	Reliability	**dependencies**	1 stations lie on every path (serial structure), but none has a failure model. Action: Fine for an ideal analysis; add failure data to find out how fragile this is.	

Performance by node

Share of the measured window each node spent busy, blocked (finished part cannot leave), down (failed) and starved (waiting for work).

Legend: green = busy, amber = blocked, red = down, grey = starved or idle.

NODE	KIND	STATE MIX	BUSY %	BLOCKED %	DOWN %	AVG / MAX WIP
green	source	-	-	-	-	-
roaster_1 bottleneck	machine		58.4	0.0	41.3	3.1 / 4
roaster_2	machine		93.4	0.0	0.4	2.7 / 4
cooling	delay		51.4	0.0	0.0	0.5 / 2
rest	delay		100.0	0.0	0.0	3.4 / 6
cupping	quality_gate		38.8	0.0	0.0	0.4 / 2
pack	batcher		19.0	0.0	0.0	1.5 / 4
shipped	sink	-	-	-	-	-
discard	sink	-	-	-	-	-

Work in process over time

Reliability and robustness

The pipeline was simulated 10 times with different random seeds. A wide interval means the result depends strongly on luck.

METRIC	MEAN	95% CONFIDENCE	STD DEV	RELATIVE
Throughput	6.74 units/h	+- 0.38	0.53	5.7%
Average WIP	11.83 units	+- 1.05	1.46	8.9%
Cycle time	5443.35 s	+- 417.45	583.59	7.7%
Arrivals lost	0.00 %	+- 0.00	0.00	0.0%

No replication deadlocked.

Failure model

NODE	MTBF	MTTR	AVAILABILITY	OBSERVED DOWN	FAILURES
roaster_1	480 min	30.0 min	94.1%	41.3%	3
roaster_2	480 min	30.0 min	94.1%	0.4%	1

Single points of failure

All units must pass through: cupping. If one of these stops, output stops.

Model validity checks

CHECK	RESULT	DETAIL
Unit conservation	PASS	created 60 = completed 44 + scrapped 4 + lost 0 + in system 12
Little's Law	WARN	L = 11.89, lambda x W = 11.17 (gap 6.1%)
Run length	WARN	window = 4 x cycle time (30x or more recommended)
Warm-up	PASS	first 60 min excluded from statistics
Deadlock-free	PASS	reference run completed

What-if analysis

Each lever was applied to one of the most constrained stations and the whole pipeline re-simulated over at least 8 hours with 3 identical random seeds per scenario. Baseline throughput: 6.8 units/h.

NODE	CHANGE	UNITS/H	CHANGE	CHANGE %
roaster_1	10% faster processing	7.0	**+0.3**	+4.2%
roaster_2	10% faster processing	7.0	**+0.2**	+3.5%
roaster_1	20% shorter repairs	6.9	**+0.1**	+2.1%
roaster_2	+2 buffer slots	6.9	**+0.1**	+1.4%
roaster_2	20% shorter repairs	6.8	**+0.0**	+0.7%
roaster_1	+2 buffer slots	6.8	**+0.0**	+0.7%
cupping	+2 buffer slots	6.8	**+0.0**	+0.0%
cupping	10% faster processing	6.8	**+0.0**	+0.0%

Best lever: 10% faster processing at 'roaster_1' -> +0.3 units/hour (4.2%), measured over 3 seeds with common random numbers.

Appendix A: node inventory

Effective parameters after node-type templates are applied. Parameters not listed use the node type's defaults.

NODE	TYPE	EFFECTIVE PARAMETERS
green	source	interarrival=exponential(420), mix={"arabica":0.6,"decaf":0.15,"robusta":0.25}, on_blocked=wait
roaster_1	machine	capacity=3, mtbf=28800, mttr=1800, process_by_type={"arabica":{"dist":"normal","mean":720,"stddev":60},"decaf":{"dist":"normal","mean":780,"stddev":60},"robusta":{"dist":"normal","mean":600,"stddev":60}}, setup=240
roaster_2	machine	capacity=3, mtbf=28800, mttr=1800, process_by_type={"arabica":{"dist":"normal","mean":720,"stddev":60},"decaf":{"dist":"normal","mean":780,"stddev":60},"robusta":{"dist":"normal","mean":600,"stddev":60}}, setup=240
cooling	delay	transit=300
rest	delay	transit=1800
cupping	quality_gate	capacity=4, max_reworks=0, pass_to=pack, process=normal(180,40), scrap_to=discard, threshold=0.08
pack	batcher	batch_size=6, capacity=12, process=240, timeout=900
shipped	sink	(defaults)
discard	sink	counts_as=scrapped

Appendix B: how the score is computed

Health score = 100 minus the sum of the penalties below, never less than 0. Grades: A 90 and above, B 80 to 89, C 70 to 79, D 60 to 69, F below 60. Penalties are capped per category so that one repeated issue cannot dominate the score. Informational findings carry no penalty.

SEVERITY	WHERE	FINDING	POINTS
High	**roaster_1**	The bottleneck is 100% utilised (busy or down). There is no slack: any disturbance directly cuts output.	-8
Medium	**roaster_1**	Down 41% of the time (3 failures).	-3
Low	**throughput**	Throughput 95% CI is +-5.7% of the mean.	-1
Low	**run length**	The measured window (7.0 h) is only 4x the cycle time.	-2
	Total deductions		**-14**
	Health score		**86**

Method: the pipeline is simulated as a discrete-event system; the bottleneck index is utilisation (busy plus down time) with small weights for starved downstream and blocked upstream neighbours. What-if results are measured, not estimated. Confidence intervals use Student's t over independent replications.